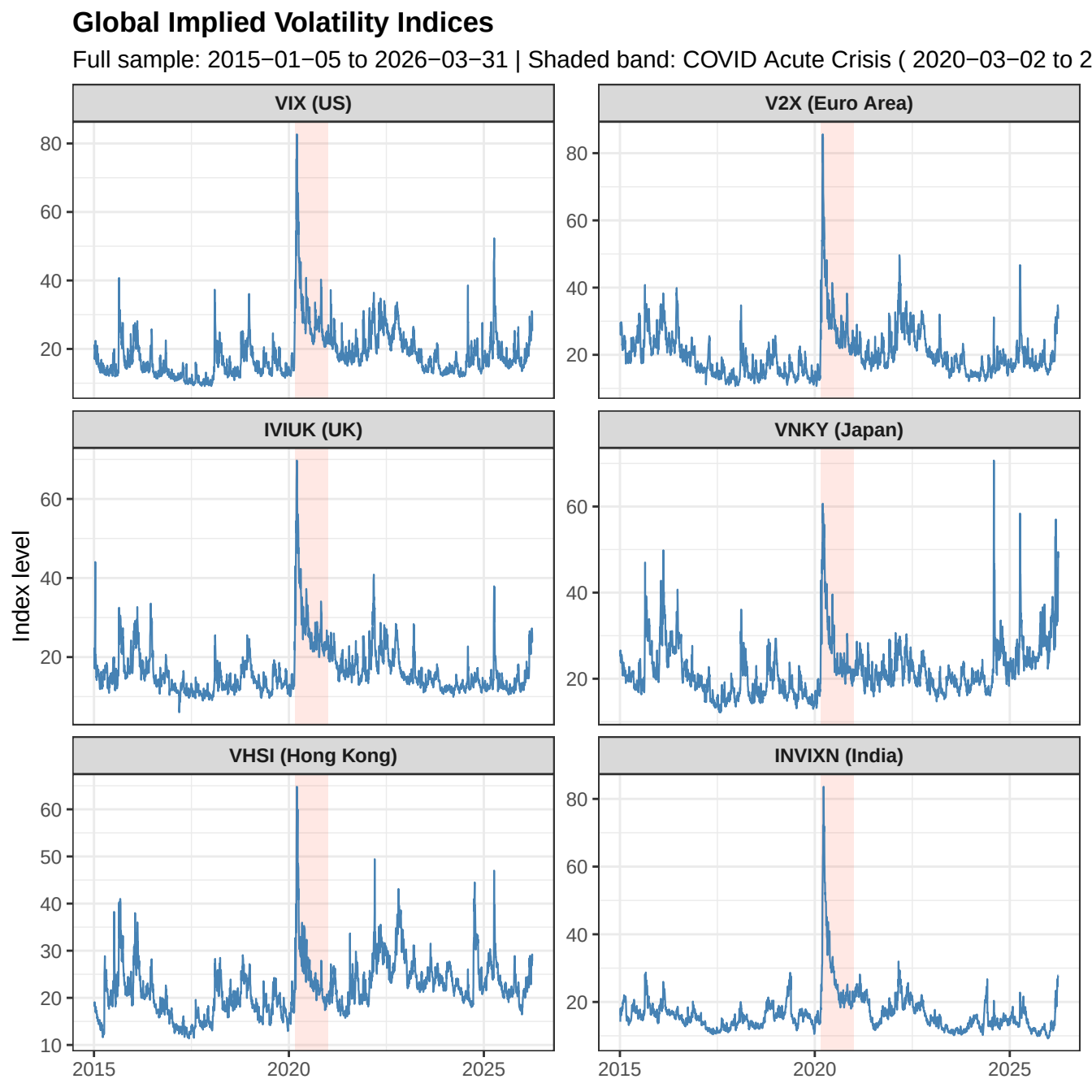
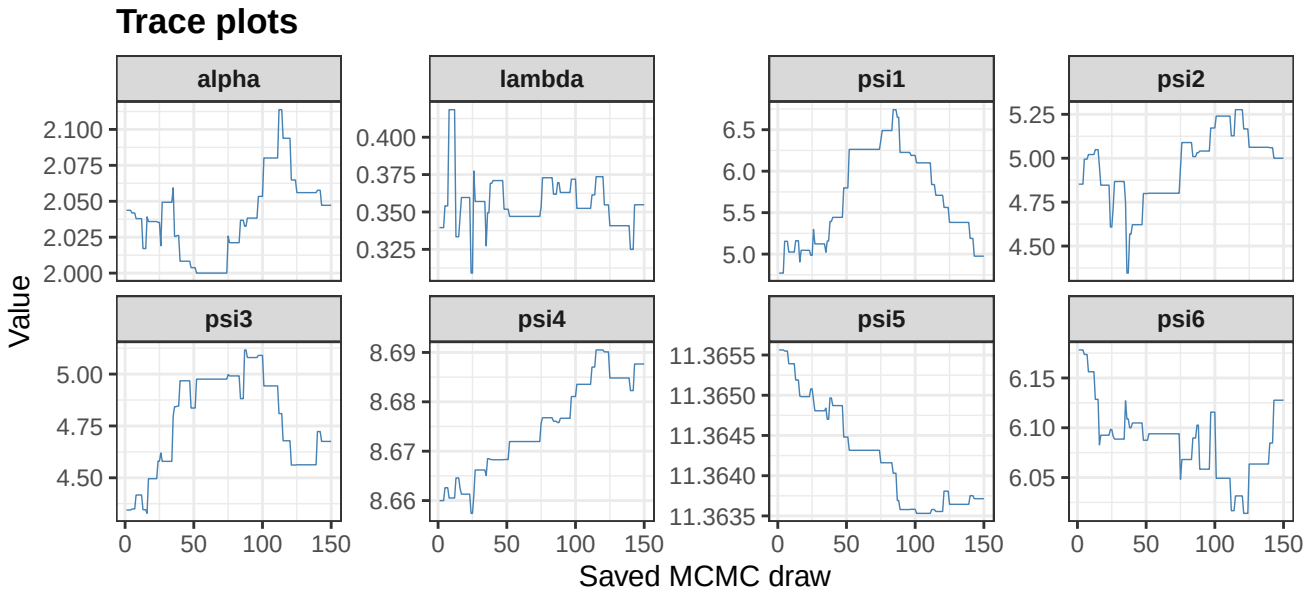
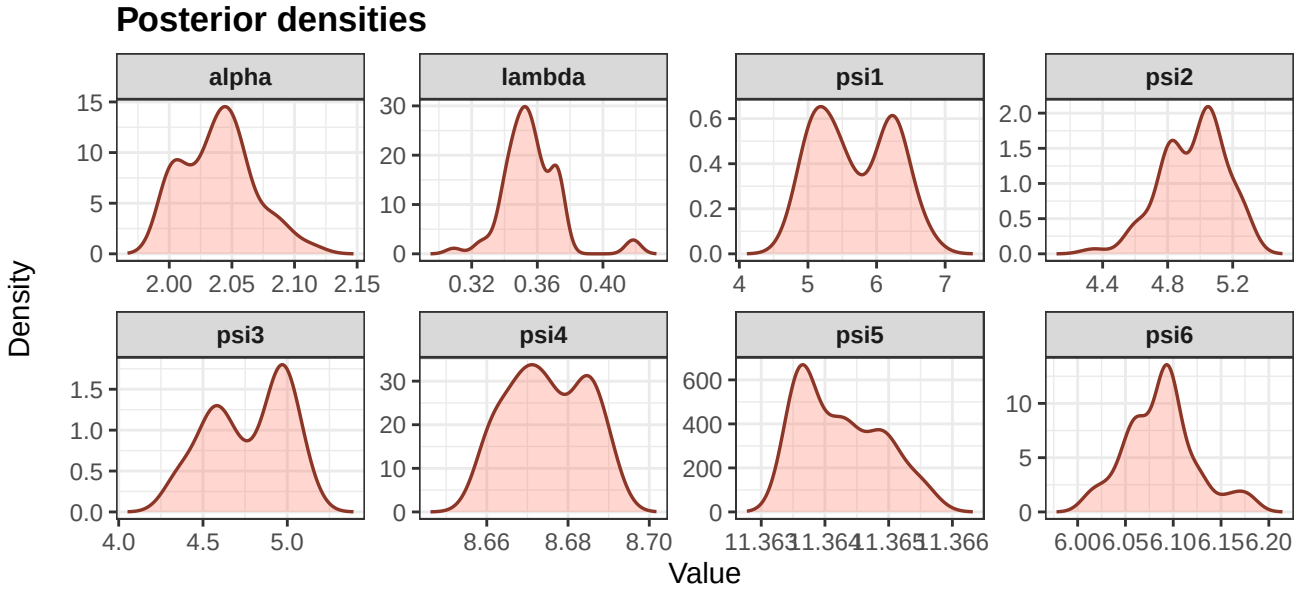
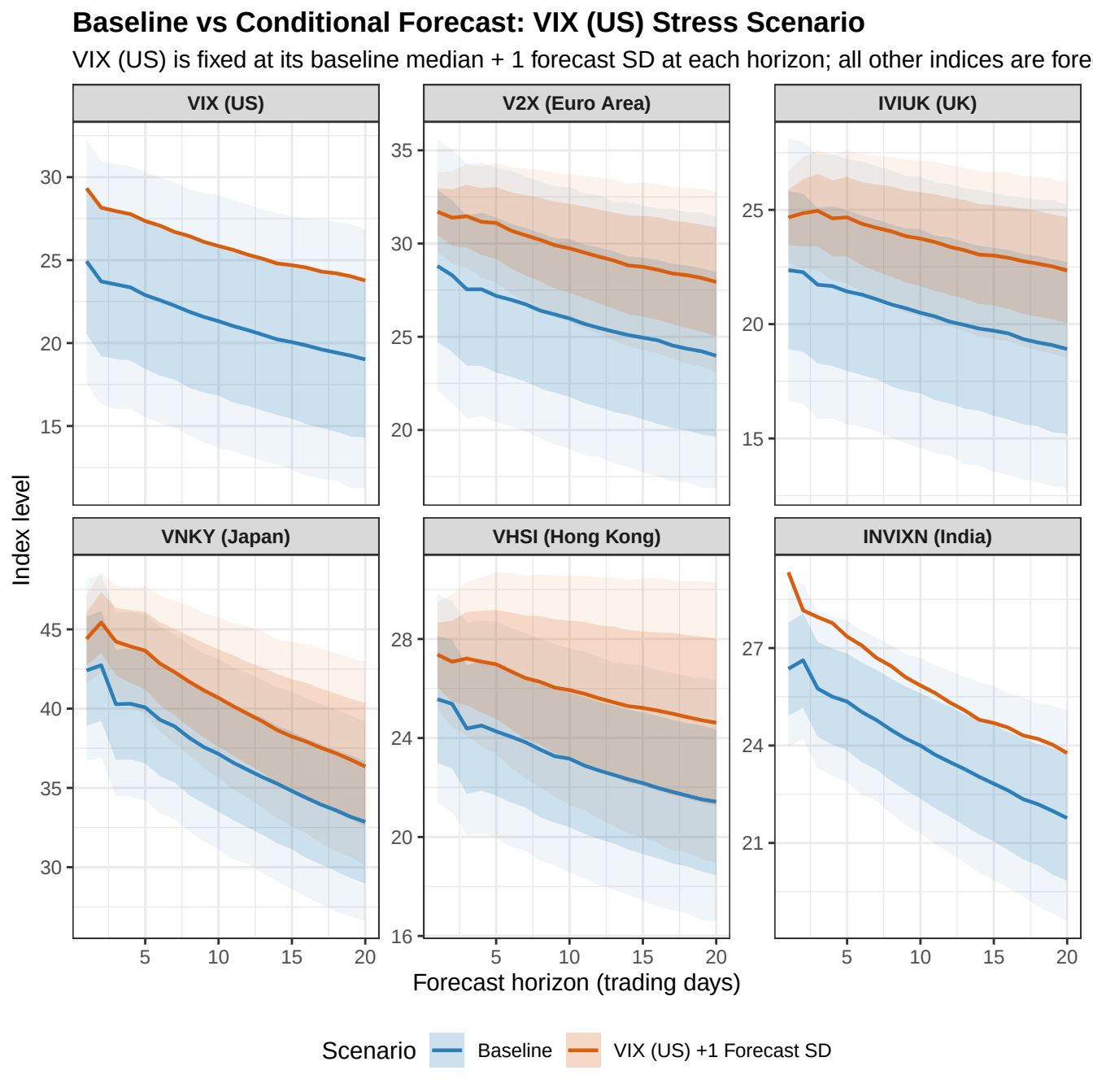

Global Implied Volatility Indices
Full sample: 2015−01−05 to 2026−03−31 | Shaded band: COVID Acute Crisis (2020−03−02 to 2
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
2015
2020
2025

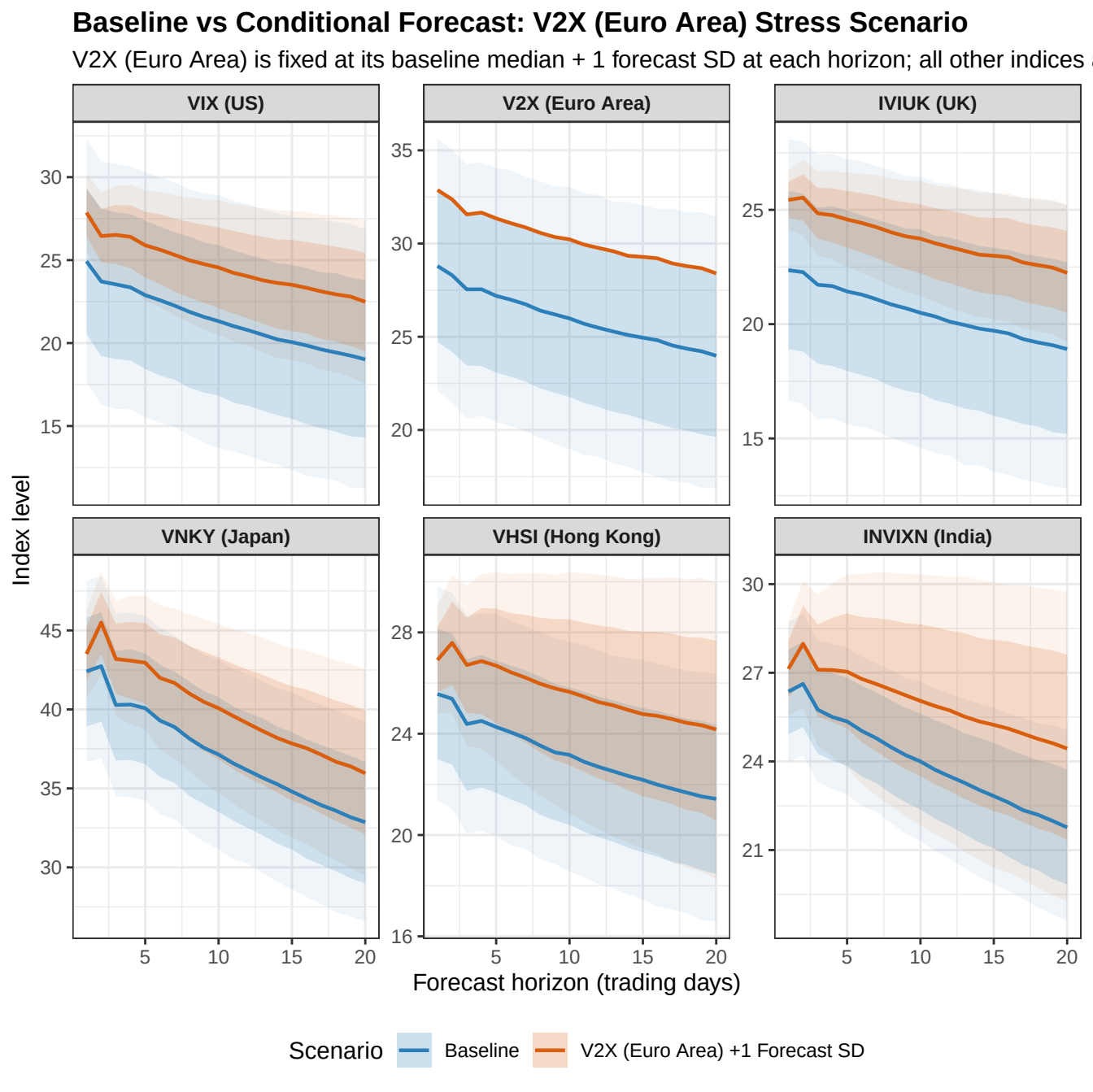
Hyperparameter Trace and Density Diagnostics − Full Sample

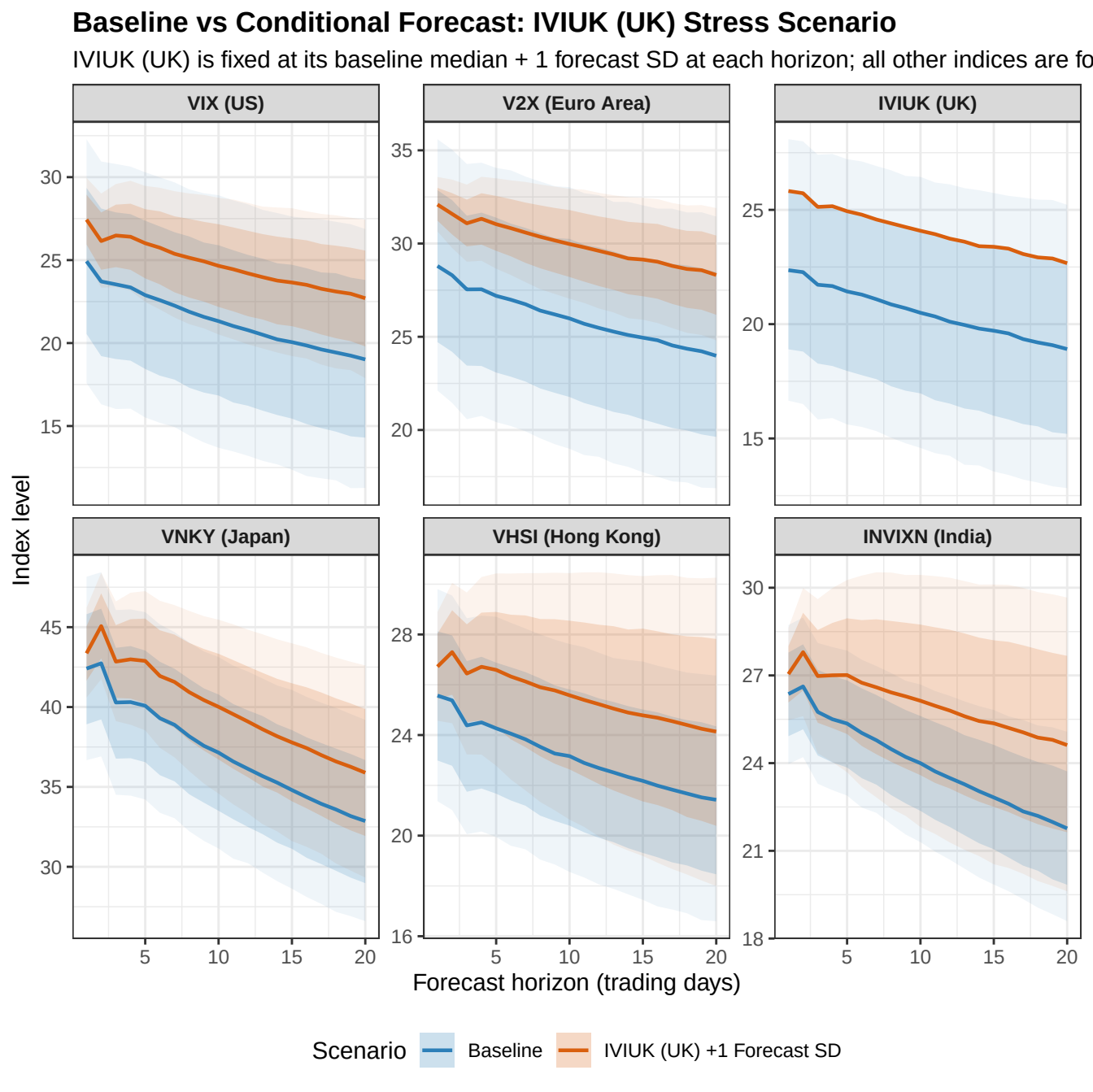
Sample: 2015−01−05 to 2026−03−31 | Geweke z outside +/−1.96 suggests non−stationarity. Multimod

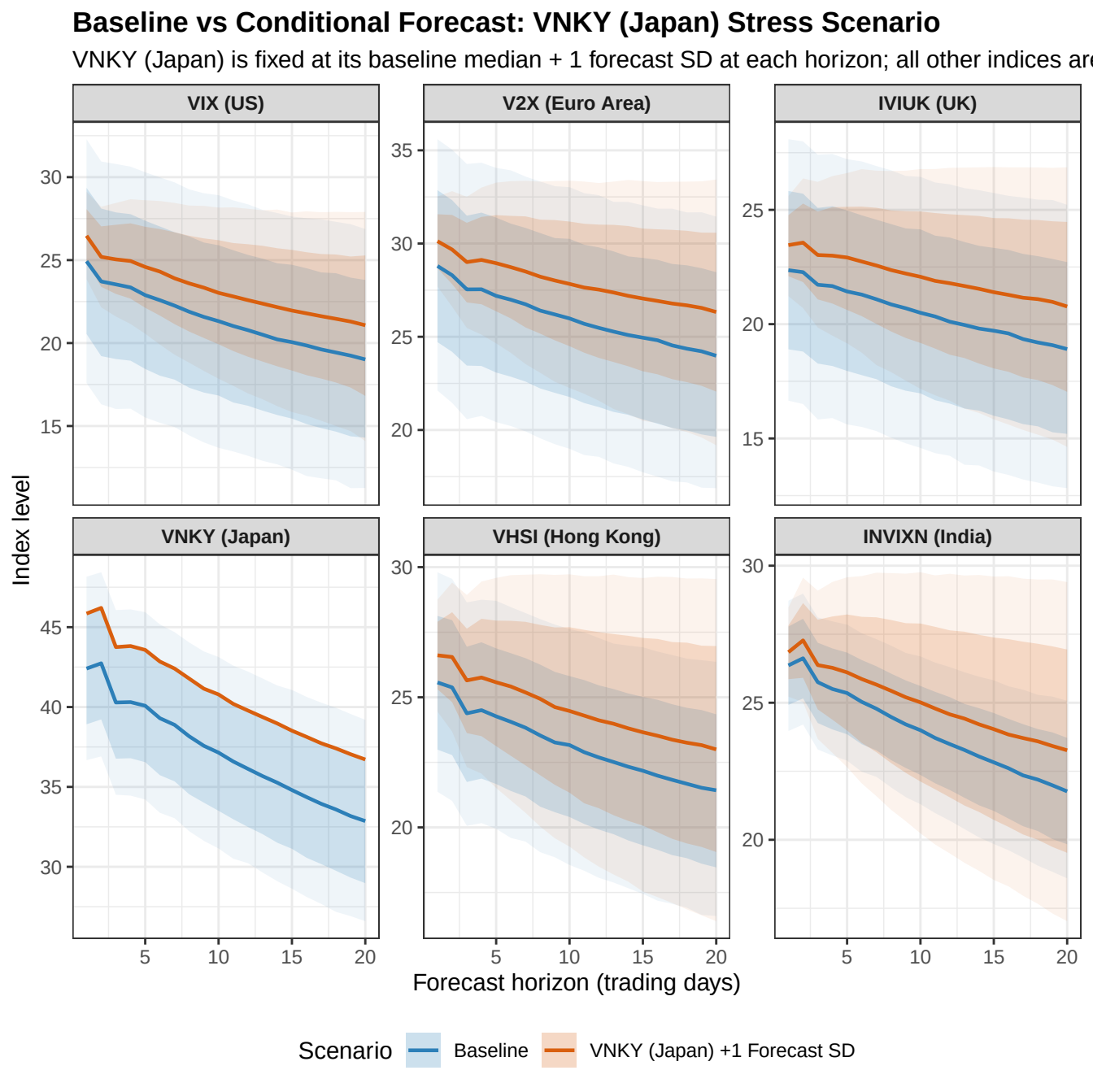


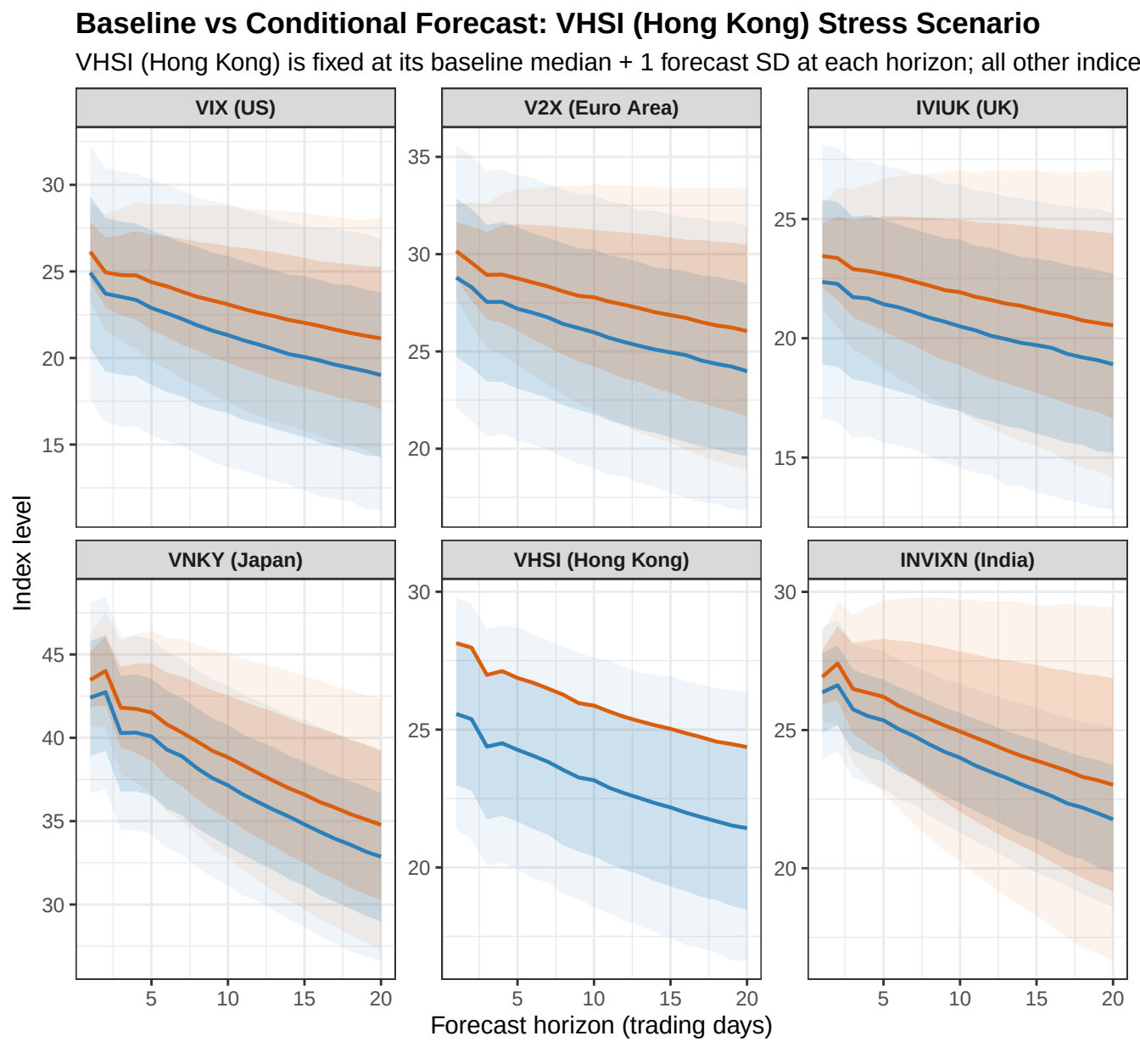


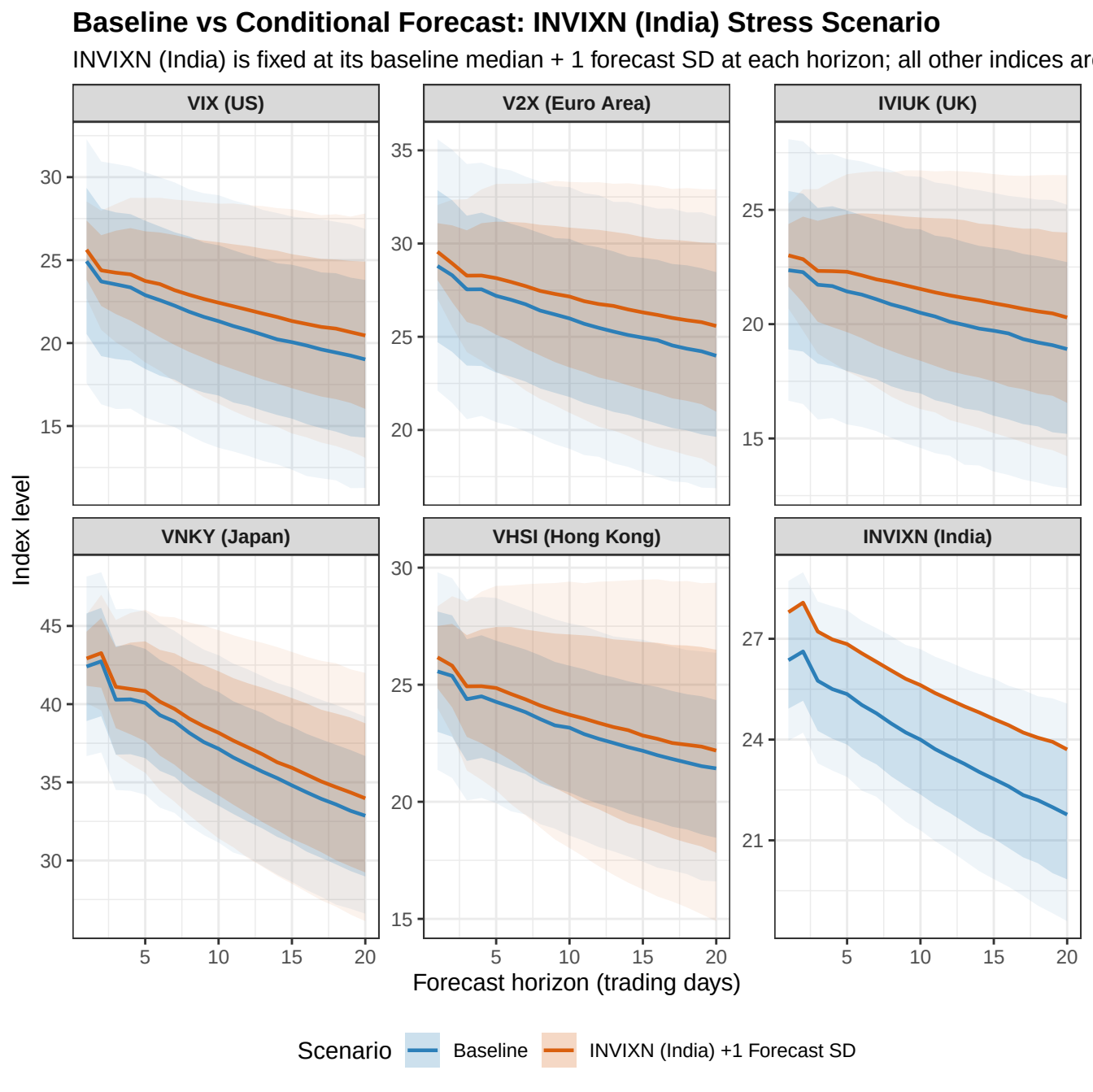

Baseline vs Conditional Forecast: VIX (US) Stress Scenario
VIX (US) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices are fore
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
VIX (US) +1 Forecast SD

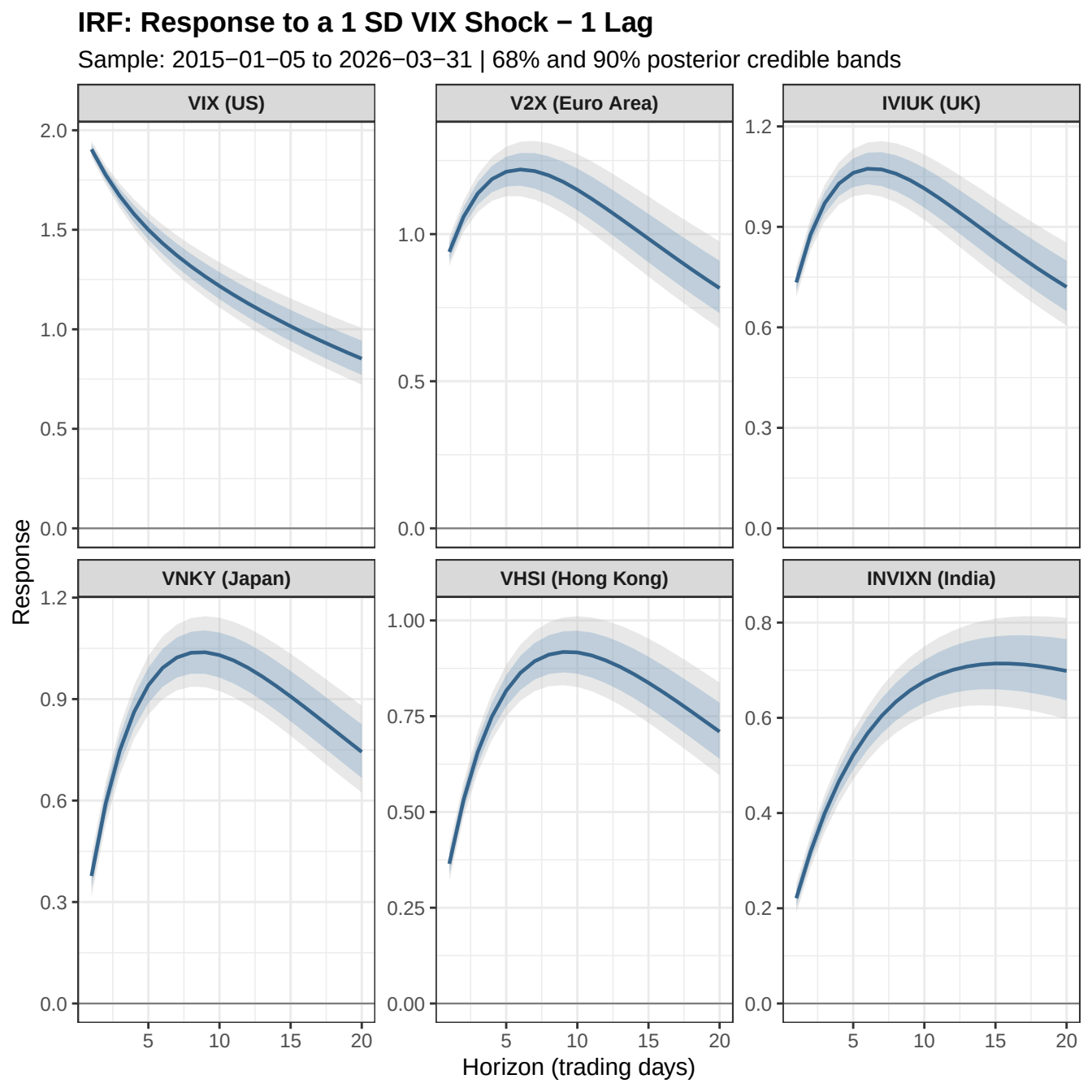

Baseline vs Conditional Forecast: V2X (Euro Area) Stress Scenario
V2X (Euro Area) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
V2X (Euro Area) +1 Forecast SD

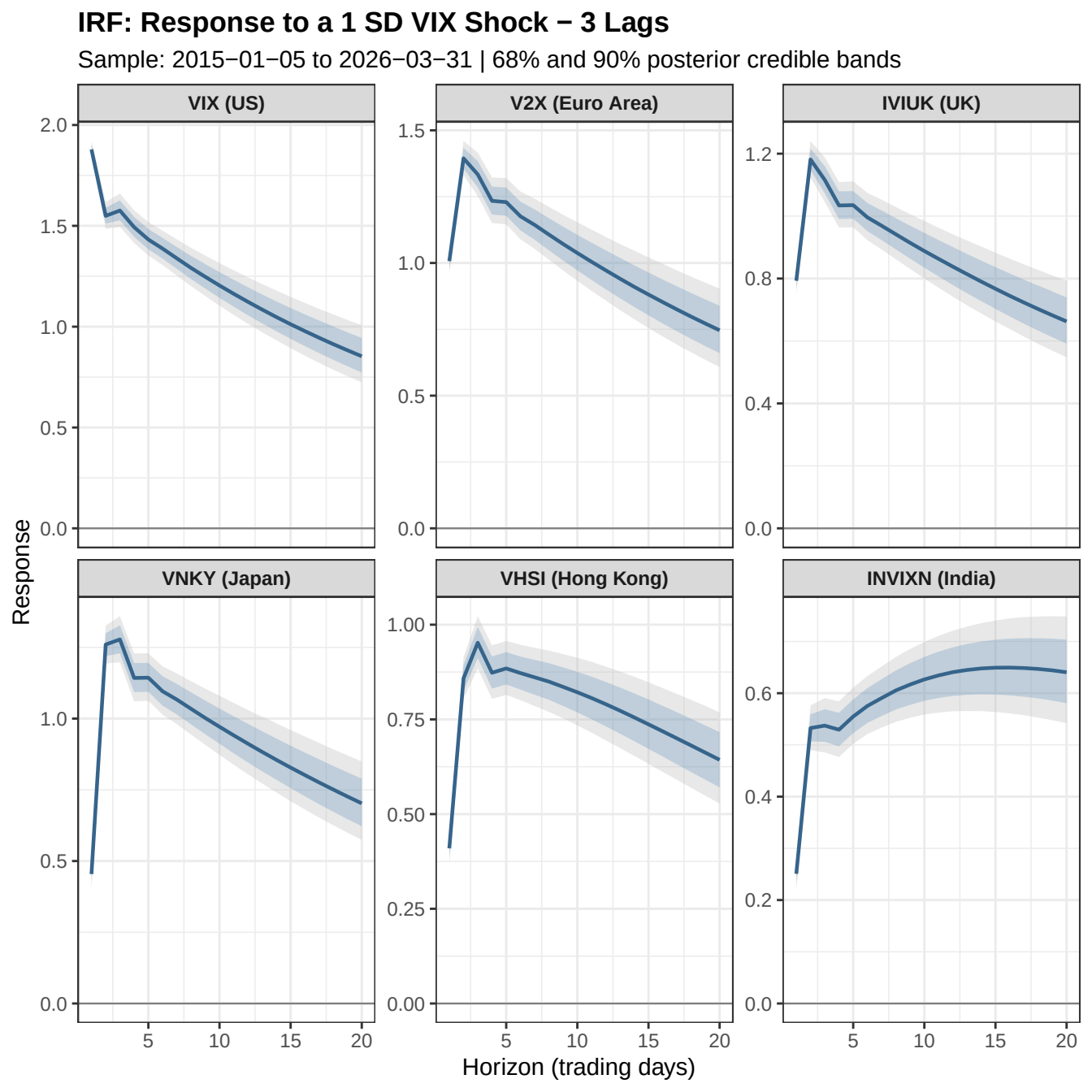

Baseline vs Conditional Forecast: IVIUK (UK) Stress Scenario
IVIUK (UK) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices are fo
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
IVIUK (UK) +1 Forecast SD

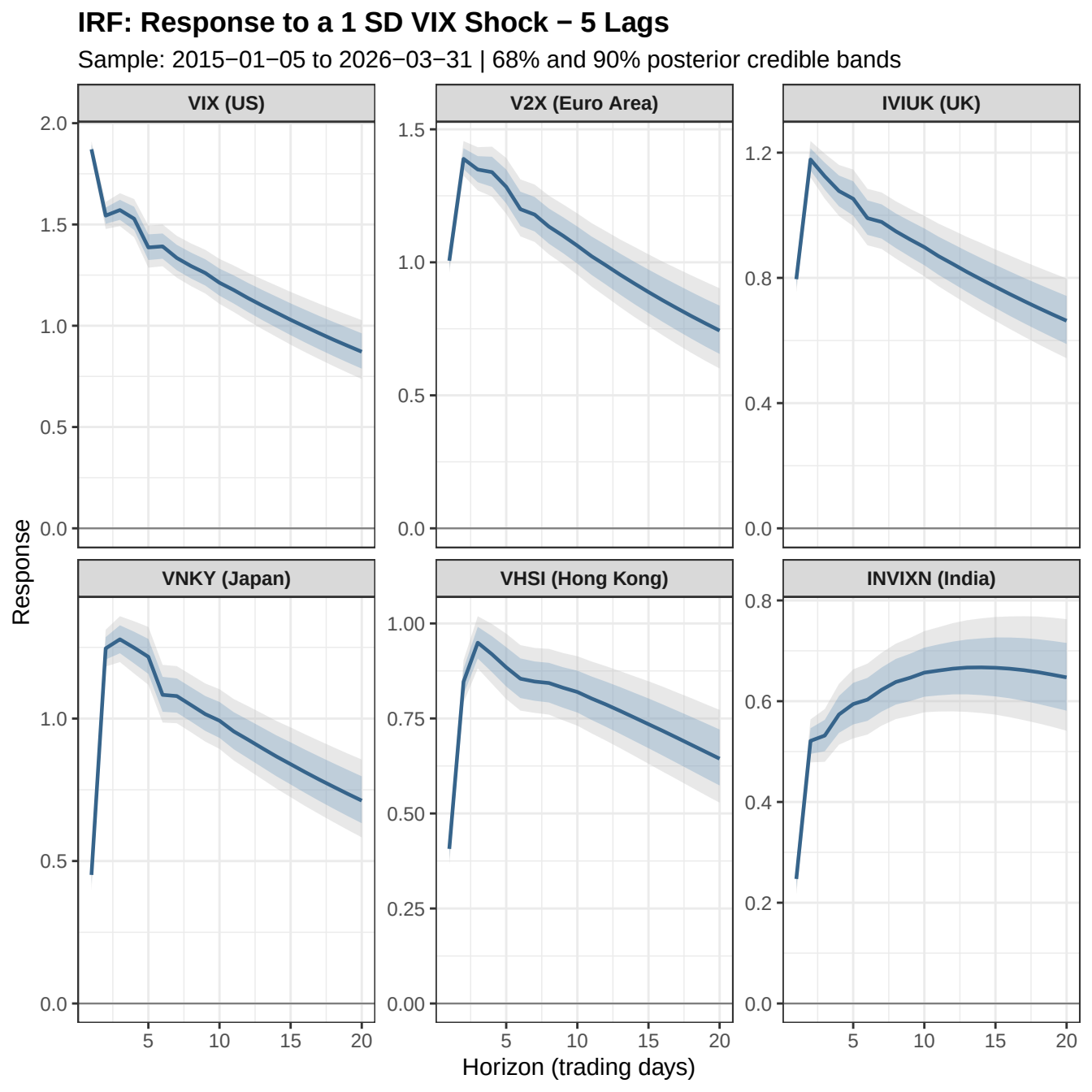

Baseline vs Conditional Forecast: VNKY (Japan) Stress Scenario
VNKY (Japan) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices are
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
VNKY (Japan) +1 Forecast SD

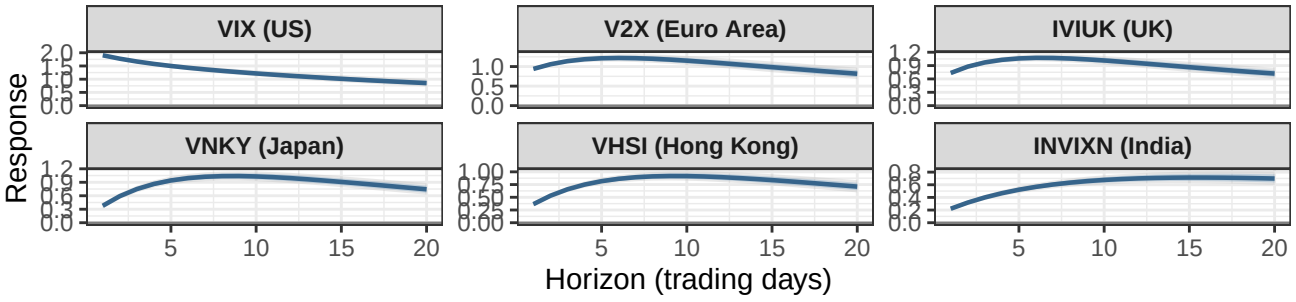
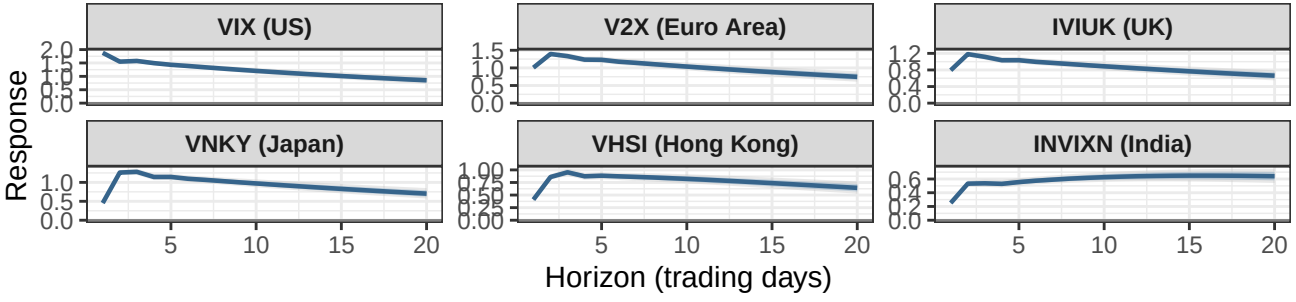
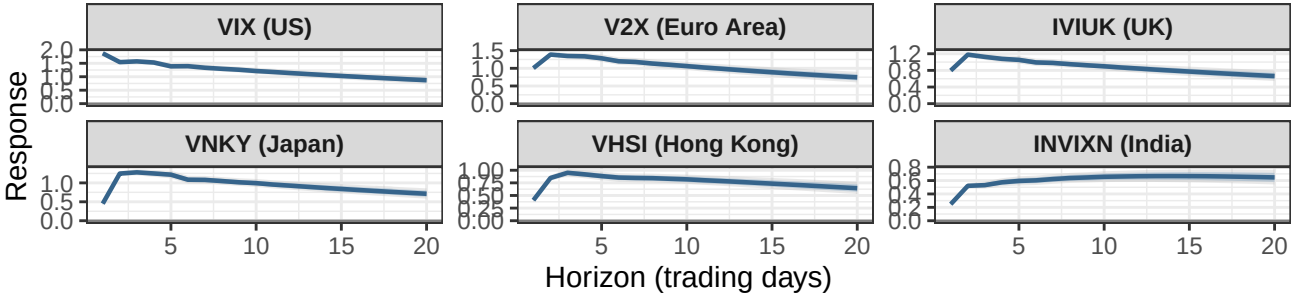

Baseline vs Conditional Forecast: VHSI (Hong Kong) Stress Scenario
VHSI (Hong Kong) is fixed at its baseline median + 1 forecast SD at each horizon; all other indice
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
VHSI (Hong Kong) +1 Forecast SD


Baseline vs Conditional Forecast: INVIXN (India) Stress Scenario
INVIXN (India) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices ar
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
INVIXN (India) +1 Forecast SD


IRF: Response to a 1 SD VIX Shock − 1 Lag
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)


IRF: Response to a 1 SD VIX Shock − 3 Lags
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)


IRF: Response to a 1 SD VIX Shock − 5 Lags
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)

Lag Sensitivity of VIX−Shock IRFs

Full sample: 2015−01−05 to 2026−03−31 | Comparing 1, 3, and 5 lags